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DISPLAY DEVICE AND OPTICAL MEMBER

Abstract

A display device includes a display panel, a transmission reflection member, a sensor, and an optical member. The transmission reflection member is provided in front of the display panel. The sensor detects an amount of light corresponding to incident light incident onto the transmission reflection member. The optical member is provided near the transmission reflection member. The optical member includes reflective surfaces continuing to each other. The optical member receives light and reflects the light by the reflective surfaces. The optical member guides the reflected light to the sensor. The reflective surfaces constitute a polygonal surface bulging out toward an opposite side of an incident side with respect to a virtual plane defined by outer contours of the reflective surfaces.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] This application is based upon and claims the benefit of priority from Japanese Patent Application No. 2024-030626, filed on Feb. 29, 2024, the entire contents of which are incorporated herein by reference.

FIELD

[0002] The present disclosure relates generally to a display device and an optical member.

BACKGROUND

[0003] A display device that includes a display panel, a transmission reflection member provided in front of the display panel, and a sensor has been known. In such a display device, an amount of light corresponding to incident light incident onto the transmission reflection member is detected by the sensor in an operation mode in which the display panel is turned off (see, for example, Patent Literature: JP 2017-183758 A).

[0004] For the above-described display device, it is desirable that the amount of light corresponding to the incident light on the transmission reflection member be appropriately detected by the sensor.

SUMMARY

[0005] A display device according to one aspect of the present disclosure includes a display panel, a transmission reflection member, a sensor, and an optical member. The transmission reflection member is provided in front of the display panel. The sensor is configured to detect an amount of light corresponding to incident light incident onto the transmission reflection member. The optical member is provided near the transmission reflection member. The optical member includes reflective surfaces continuing to each other. The optical member is configured to receive light, reflect the light by the reflective surfaces, and guide the reflected light to the sensor. The reflective surfaces constitute a polygonal surface bulging out toward an opposite side of an incident side with respect to a virtual plane defined by outer contours of the reflective surfaces.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0006] FIG. 1 is a diagram schematically illustrating a vehicle on which a display system including a display device according to an embodiment is installed;

[0007] FIG. 2 is a perspective view illustrating an external configuration of the display device according to the embodiment;

[0008] FIG. 3 is a front view illustrating the external configuration of the display device according to the embodiment;

[0009] FIG. 4 is a cross-sectional view illustrating a configuration of the display device according to the embodiment;

[0010] FIG. 5 is a perspective view illustrating a configuration of an optical member in the embodiment;

[0011] FIG. 6 is a vertical cross-sectional view illustrating the configuration of the optical member in the embodiment;

[0012] FIG. 7 is a cross-sectional view in a planar direction, illustrating the configuration of the

optical member in the embodiment;

[0013] FIG. **8** is a perspective view illustrating an operation of the optical member in the embodiment;

[0014] FIG. **9** is a diagram illustrating vertical angle dependency of sensor sensitivity in the embodiment;

[0015] FIG. **10** is a perspective view illustrating the operation of the optical member in the embodiment;

[0016] FIG. **11** is a diagram illustrating horizontal angle dependency of sensor sensitivity in the embodiment;

[0017] FIG. **12** is a perspective view illustrating a configuration of an optical member in a modification of the embodiment;

[0018] FIG. **13** is a vertical cross-sectional view illustrating the configuration of the optical member in the modification of the embodiment; and

[0019] FIG. **14** is a cross-sectional view in a planar direction, illustrating the configuration of the optical member in the modification of the embodiment.

DETAILED DESCRIPTION

[0020] Hereinafter, an embodiment of a display device according to the present disclosure will be described with reference to the drawings.

Embodiment

[0021] While a display device according to an embodiment includes a display panel, a transmission reflection member provided in front of the display panel, and a sensor, and detects an amount of light corresponding to incident light incident onto the transmission reflection member by the sensor, contrivance for appropriately detecting the amount of light by the sensor has been conducted.

[0022] A display system **3** including a display device **1** according to the embodiment can be installed in a vehicle **5**, as illustrated in FIG. **1**. In one example, the display device **1** is an electronic mirror, and the display system **3** is an electronic mirror system.

[0023] FIG. **1** is a diagram illustrating the vehicle **5** on which the display system **3** is installed. The display system **3** includes the display device **1** and an imaging device **2**.

[0024] The imaging device **2** is an in-vehicle camera installed in the vehicle **5**, and is installed outside or inside a vehicle body **6**. The imaging device **2** may be installed at an end part on the rear side of the vehicle body **6** to image the rear, may be installed at an end part near the door of the vehicle body **6** to image the side, or may be installed at an end part on the front side of the vehicle body **6** to image the front.

[0025] The display device **1** is provided in a vehicle compartment **7**. The display device **1** is, for example, an electronic mirror. The display device **1** has a front surface **1a** and can display an image acquired by the imaging device **2** on the front surface **1a**. The display device **1** is configured to be switchable between a display mode and a mirror mode. The display mode is a mode in which a display panel in the display device **1** displays an image captured by the imaging device **2** and thereby the display device **1** functions as a display. The mirror mode is a mode in which the display panel in the display device **1** is turned off and thereby the display device **1** functions as a mirror.

[0026] In a case where the display device **1** is an electronic mirror used for rear visual recognition, the display device **1** may be installed in the form of a rearview mirror. In this case, the front surface **1a** may face the vehicle compartment **7**, and the shape of the front surface **1a** may be the shape of the mirror surface of the rearview mirror. In a case where the display device **1** is an electronic mirror for side visual recognition, the display device **1** may be installed in the form of a door mirror (for example, a door mirror **61**). In this case, the front surface **1a** may face the rear of the vehicle body **6**, and the shape of the front surface **1a** may be the shape of the mirror surface of the door mirror. In a case where the display device **1** is an electronic mirror for front visual recognition, the display device **1** may be installed in the form of a vehicle-installed display device (for example, a

display device **71**). In this case, and the front surface **1a** may face the vehicle compartment **7**, and the shape of the front surface **1a** may be the shape of a display panel of the vehicle-installed display device (**71**).

[0027] In the example illustrated in FIG. **1**, the imaging device **2** is installed at a rear end part **6a** of the vehicle body **6**, and the display device **1** is applied to an electronic mirror system for rear visual recognition. The electronic mirror system for rear visual recognition is also called an electronic rearview mirror. The imaging device **2** acquires an image behind the vehicle body. The display device **1** can display an image of the rear side of the vehicle body, in which the image is captured by the imaging device **2**.

[0028] When the display device **1** is an electronic mirror for rear visual recognition, the display device **1** can be configured as illustrated in FIGS. **2** and **3**. FIG. **2** is a perspective view illustrating an external configuration of the display device **1**. FIG. **3** is a front view illustrating an external configuration of the display device **1**.

[0029] The display device **1** may have a back surface **1b** on the opposite side of the front surface **1a**, and may be fixed to the vehicle body of the vehicle **5** via a fixing member **20** on the back surface **1b** side. The display device **1** includes a housing **10**, a transmission reflection member **11**, a display panel **12**, and an optical member **100**.

[0030] The housing **10** has a substantially rectangular parallelepiped appearance shape. Hereinafter, a direction perpendicular to the front surface **1a** of the display device **1** is referred to as the X direction, a longitudinal direction of the housing **10** is referred to as the Y direction, and a direction perpendicular to the X direction and the Y direction is referred to as the Z direction. The housing **10** has an opening **10a** on the $-X$ side, an opening **10b** on the $+X$ side, and an opening **10c** on the $-Z$ side.

[0031] One end of an arm **21** in the fixing member **20** is rotatably inserted into the opening **10b**. The other end of the arm **21** is connected to an installation member **22** in the fixing member **20**. The installation member **22** may be attached inside the vehicle body **6**. Accordingly, the housing **10** can be fixed to the vehicle body **6**.

[0032] The transmission reflection member **11** is provided on the $-X$ side of the display panel **12** and is fixed to the housing **10** from the $-X$ side. The transmission reflection member **11** extends in a plate shape in the YZ direction and has a substantially rectangular shape or a substantially inverted isosceles trapezoidal shape in YZ plane view. In the transmission reflection member **11**, corners of a substantially rectangular shape or a substantially inverted isosceles trapezoidal shape may be rounded. The transmission reflection member **11** has a YZ dimension corresponding to the opening **10a**. The Y width of the transmission reflection member **11** is slightly larger than the Y width of the opening **10a**. The Z width of the transmission reflection member **11** is slightly larger than the Z width of the opening **10a**. Thus, the transmission reflection member **11** can close the opening **10a** from the $-X$ side.

[0033] The display panel **12** is provided on the $+X$ side of the transmission reflection member **11** and is housed in the housing **10**. The display panel **12** extends in a plate shape in the YZ direction and has a substantially rectangular shape in YZ plane view. The display panel **12** has a YZ dimension corresponding to the opening **10a**. The Y width of the display panel **12** is slightly larger than the Y width of the opening **10a**. The Z width of the display panel **12** is slightly larger than the Z width of the opening **10a**. Consequently, the display panel **12** can close the opening **10a** from the $+X$ side.

[0034] The optical member **100** is provided near the transmission reflection member **11**. The optical member **100** can be provided on the $-Z$ side of the transmission reflection member **11**. In the optical member **100**, an incident surface **101** protrudes from the opening **10c** to the $-Z$ side. The incident surface **101** faces the $-X$ side and receives light incident from the $-X$ side. Light from a headlight of another vehicle, which is present behind the vehicle **5**, can be incident onto an incident surface **101a**. The optical member **100** may be made of material with translucency. The

optical member **100** may be made of glass, quartz, transparent plastic, or the like.

[0035] FIG. **4** is a cross-sectional view illustrating a configuration of the display device. FIG. **4** illustrates a cross section of FIG. **3** taken along line A-A. As illustrated in FIG. **4**, the display device **1** further includes a substrate **13**, a substrate **14**, a flexible flat cable (FFC) **15**, and a sensor **16**.

[0036] The substrate **13** is provided on the +X side of the display panel **12** and is housed in the housing **10**. The substrate **13** extends in a plate shape in the YZ direction and has a substantially rectangular shape in YZ plane view. The substrate **13** may have a smaller dimension than that of the display panel **12** in YZ plane view. The -Z side end part of the substrate **13** may be located closer to the +Z side than the -Z side end part of the display panel **12**. A control circuit element **13a** may be installed in the +X side surface of the substrate **13**. The control circuit element **13a** may include a circuit for controlling the reflectance of the transmission reflection member **11**.

[0037] The substrate **14** is provided between the substrate **13** and the optical member **100**. The substrate **14** is provided on the -Z side of the substrate **13**, is provided on the +Z side of the optical member **100**, and is housed in the housing **10**. The substrate **13** extends in the plate shape in the XY direction and has a substantially rectangular shape in XY plan view. The substrate **14** may be larger in size than the optical member **100** in XY plane view. A control circuit element **14a** may be installed in the +Z side surface of the substrate **14**. The control circuit element **14a** may include a circuit for controlling the reflectance of the transmission reflection member **11**.

[0038] The FFC **15** may be a bendable film-shaped cable. The FFC **15** is provided between the control circuit element **13a** and the control circuit element **14a**. One end of the FFC **15** is electrically connected to the control circuit element **13a**, and the other end thereof is electrically connected to the control circuit element **14a**.

[0039] The sensor **16** is installed in the -Z side surface of the substrate **14** and is housed in the housing **10**. The sensor **16** can detect the amount of light. The sensor **16** is an illuminance sensor, and may be, for example, a silicon photosensor. In the sensor **16**, a light reception region **161** (refer to FIG. **10**) has an area of A1. The sensor **16** can accumulate a charge corresponding to light received on the light reception surface and output a signal corresponding to the accumulated charge as a detection result. The sensor **16** may be electrically connected to the control circuit element **14a** via wiring in the substrate **14**. With this configuration, the sensor **16** can transmit the detection result to the control circuit element **13a** via the wiring in the substrate **14**, the control circuit element **14a**, and the FFC **15**. With this configuration, the control circuit element **13a** can adjust the reflectance of the transmission reflection member **11** to reflectance corresponding to the amount of light detected by the sensor **16**.

[0040] In one example, the control circuit element **13a** controls, in a normal state, the reflectance of the transmission reflection member **11** to be RR1 in a state where the display device **1** operates in the mirror mode, and then controls the reflectance of the transmission reflection member **11** to be RR2 (<RR1) when the amount of light detected by the sensor **16** exceeds a threshold amount of light. With this processing, if the front surface **1a** receives light from a headlight of another vehicle behind the vehicle **5**, the intensity of the light that has been reflected by the front surface **1a** and visually recognized by a passenger of the vehicle **5** can be suppressed. Therefore, the convenience for the passenger of the vehicle **5** can be improved.

[0041] The optical member **100** is provided on the -Z side of the sensor **16**. As illustrated in FIGS. **5** to **7**, the optical member **100** has plural reflective surfaces **102a** to **102i** continuing to each other. FIG. **5** is a perspective view illustrating a configuration of the optical member **100**. FIG. **6** is an XZ cross-sectional view illustrating the configuration of the optical member **100**, and is an XZ enlarged cross-sectional view of a section B in FIG. **4** as viewed from the -Y direction. FIG. **7** is an XY cross-sectional view illustrating the configuration of the optical member **100**, and illustrates a cross section of FIG. **6** taken along line C-C.

[0042] The optical member **100** includes a reflective surface group **102** and an emitting surface **103**

in addition to the incident surface **101**. The incident surface **101** is provided at positions on the $-X$ side and the $-Z$ side of the optical member **100**, is exposed as an outer surface of the optical member **100**, and faces the $-X$ side. The reflective surface group **102** is provided on the $+X$ side of the incident surface **101**. The reflective surface group **102** is provided at positions on the $+X$ side and the $-Z$ side in the optical member **100** and faces a direction inclining to the $-X$ side and the $+Z$ side in the optical member **100**. The emitting surface **103** is provided on the $+Z$ side of the reflective surface group **102**. The emitting surface **103** is provided at a position on the $+Z$ side in the optical member **100**, is exposed as an outer surface of the optical member **100**, and faces the $+Z$ side.

[0043] The optical member **100** receives light incident from the $-X$ side by the incident surface **101** and reflects the received light by the reflective surface group **102**. The optical member **100** then emits the reflected light from the emitting surface **103** and guides the reflected light to the sensor **16**. The reflective surface group **102** is provided between the incident surface **101** and the emitting surface **103** in the optical path.

[0044] The reflective surface group **102** includes the reflective surfaces **102a** to **102i** continuing to each other. The reflective surfaces **102a** to **102i** constitute a polygonal surface bulging out toward the opposite side of the incident side with respect to a virtual plane VP. The virtual plane VP is defined by the outer contours of the reflective surfaces **102a** to **102i** (refer to FIG. 5). With this configuration, the optical member **100** can guide the incident light from the $-X$ side to the sensor **16** in a wide angle range, and can expand the angle range in which the sensor **16** can detect the amount of light.

[0045] The incident surface **101** includes convex surfaces **101a_1** to **101a_5**, **101b_1** to **101b_5**, **101c_1** to **101c_5**, and **101d_1** to **104d_5**. Each of the convex surfaces **101a_1** to **104d_5** includes a curved surface that is convex to the $-X$ side. With this structure, the optical member **100** can receive light in a wide angle range on the incident surface **101**.

[0046] In the present specification, regarding an incident angle of light onto the incident surface **101**, an incident angle in the vertical direction and an incident angle in the horizontal direction are each considered. It is assumed that the incident surface **101** is a surface extending substantially in the YZ direction, and the $-X$ direction is a normal direction of the incident surface **101**. The incident angle in the vertical direction with respect to the incident surface **101** is defined such that an angle formed clockwise in the $-X$ direction is defined as a positive angle and an angle formed counterclockwise is defined as a negative angle when viewed from the $-Y$ direction. The incident angle in the horizontal direction with respect to the incident surface **101** is defined such that an angle formed clockwise in the $-X$ direction is defined as a positive angle and an angle formed counterclockwise is defined as a negative angle when viewed from the $+Z$ direction.

[0047] The reflective surfaces **102a** to **102i** included in the reflective surface group **102** incline and continue to each other, thereby constituting a polygonal surface.

[0048] In the present specification, the inclination angle in the vertical direction and the inclination angle in the horizontal direction are each considered for the inclination angles of the two reflective surfaces. An inclination angle of a first reflective surface in the vertical direction with respect to a second reflective surface is defined such that an angle formed clockwise with respect to the second reflective surface is defined as a positive angle and an angle formed counterclockwise with respect to the second reflective surface is defined as a negative angle when viewed from the $-Y$ direction. An inclination angle of the first reflective surface in the horizontal direction with respect to the second reflective surface is defined such that an angle formed clockwise with respect to the second reflective surface is defined as a positive angle and an angle formed counterclockwise with respect to the second reflective surface is defined as a negative angle when viewed from the $+Z$ direction.

[0049] The reflective surfaces **102a** to **102i** are obtained by dividing the virtual plane VP by 3×3 and deforming the virtual plane VP so as to bulge out toward the opposite side of the incident side while keeping a state where 3×3 reflective surfaces are connected to each other. The reflective

surfaces **102a** to **102i** are connected to each other via straight sides.

[0050] The reflective surfaces **102a** to **102i** incline at an angle to each other. Among the reflective surfaces **102a** to **102i**, the reflective surface **102b** extends along the virtual plane VP, and the reflective surfaces **102a** and **102c** to **102i** extend obliquely to the virtual plane VP.

[0051] The reflective surfaces **102a** to **102i** may have different incident angles of incident light to be respectively reflected. The reflective surface **102b** is allotted incident light having a relatively small incident angle. The reflective surfaces **102a** and **102c** are allotted incident light having a relatively large incident angle in the vertical direction on the positive side and the negative side, respectively. The reflective surfaces **102e** and **102h** are allotted incident light having a relatively large incident angle in the horizontal direction on the positive side and the negative side, respectively. The reflective surfaces **102d** and **102g** are allotted incident light having a relatively large incident angle in the vertical direction on the positive side and a relatively large incident angle in the horizontal direction on the positive side and the negative side, respectively. The reflective surfaces **102i** and **102f** are allotted incident light having a relatively large incident angle in the vertical direction on the negative side and a relatively large incident angle in the horizontal direction on the positive side and the negative side, respectively.

[0052] Each of the reflective surfaces **102a** to **102i** has a substantially rectangular shape in plan view. The areas of the respective reflective surfaces **102a** to **102i** may be equal to each other. The areas of the respective reflective surfaces **102a** to **102i** may be equal to or larger than the area A1 of the light reception region **161** of the sensor **16**. The area of each of the reflective surfaces **102a** to **102i** is preferably $1.5\text{ mm} \times 1.5\text{ mm} = 2.25\text{ mm}^2$ or more. The area of each of the reflective surfaces **102a** to **102i** is more preferably $3\text{ mm} \times 3\text{ mm} = 9\text{ mm}^2$ or more.

[0053] The reflective surface **102b** is provided at the center of the reflective surfaces **102a** to **102i**. The reflective surface **102b** is connected to the reflective surface **102a** at the side on the $-Z$ side, connected to the reflective surface **102c** at the side on the $+Z$ side, connected to the reflective surface **102e** at the side on the $-Y$ side, and connected to the reflective surface **102h** at the side on the $+Y$ side. The corner of the reflective surface **102b** on the $-Y$ side and the $-Z$ side is in contact with the corner of the reflective surface **102d**, the corner of the reflective surface **102b** on the $-Y$ side and the $+Z$ side is in contact with the corner of the reflective surface **102f**, the corner of the reflective surface **102b** on the $+Y$ side and the $-Z$ side is in contact with the corner of the reflective surface **102g**, and the corner of the reflective surface **102b** on the $+Y$ side and the $+Z$ side is in contact with the corner of the reflective surface **102i**. The reflective surface **102b** may be substantially parallel to the virtual plane VP (refer to FIGS. 6 and 7). The area of the reflective surface **102b** may be smaller than the area of each of the other reflective surfaces **102a** and **102c** to **102i**.

[0054] The reflective surface **102a** is provided on the $-Z$ side of the reflective surface **102b** and is connected to the reflective surface **102b** at the side on the $+Z$ side. The reflective surface **102a** inclines with respect to the reflective surface **102b** such that the $-Z$ side is closer to the virtual plane VP than the $+Z$ side. An angle between the reflective surface **102a** and the virtual plane VP corresponds to the incident angle of light to the incident surface **101**. An angle between the reflective surface **102a** and the reflective surface **102b** corresponds to the incident angle of light to the incident surface **101**. An inclination angle θ_{ah} of the reflective surface **102a** in the horizontal direction with respect to the reflective surface **102b** corresponds to an incident angle of light in the horizontal direction to be covered by the reflective surface **102a**. An inclination angle θ_{av} of the reflective surface **102a** in the vertical direction with respect to the reflective surface **102b** corresponds to an incident angle of light in the vertical direction to be covered by the reflective surface **102a**.

[0055] The reflective surface **102a** is allotted incident light having a relatively large incident angle in the vertical direction on the positive side. The angle between the reflective surface **102a** and the reflective surface **102b** may be about $\frac{1}{4}$ of the incident angle of the incident light to be covered by

the reflective surface **102a**. If the optical member **100** is required to cover incident light having an incident angle of -10° or more and $+10^\circ$ or less in the vertical direction, the inclination angle θ_{av} of the reflective surface **102a** in the vertical direction with respect to the reflective surface **102b** may be approximately 2.5° . The inclination angle θ_{ah} (refer to FIG. 6) of the reflective surface **102a** in the horizontal direction with respect to the reflective surface **102b** may be substantially 0. [0056] The reflective surface **102c** is provided on the +Z side of the reflective surface **102b**, and is connected to the reflective surface **102b** at the side on the -Z side. The reflective surface **102c** inclines with respect to the reflective surface **102b** such that the +Z side is closer to the virtual plane VP than the -Z side. An angle between the reflective surface **102c** and the virtual plane VP corresponds to the incident angle of light to the incident surface **101**. An angle between the reflective surface **102c** and the reflective surface **102b** corresponds to the incident angle of light to the incident surface **101**. An inclination angle θ_{ch} of the reflective surface **102c** in the horizontal direction with respect to the reflective surface **102b** corresponds to an incident angle of light in the horizontal direction to be covered by the reflective surface **102c**. An inclination angle θ_{cv} of the reflective surface **102c** in the vertical direction with respect to the reflective surface **102b** corresponds to an incident angle of light in the vertical direction to be covered by the reflective surface **102c**.

[0057] The reflective surface **102c** is allotted incident light having a relatively large incident angle in the vertical direction on the positive side. The angle between the reflective surface **102c** and the reflective surface **102b** may be about $\frac{1}{4}$ of the incident angle of the incident light to be covered by the reflective surface **102c**. If the optical member **100** is required to cover incident light having an incident angle of -10° or more and $+10^\circ$ or less in the vertical direction, the inclination angle θ_{cv} (refer to FIG. 6) of the reflective surface **102c** in the vertical direction with respect to the reflective surface **102b** may be approximately -2.5° . The inclination angle θ_{ch} of the reflective surface **102c** in the horizontal direction with respect to the reflective surface **102b** may be substantially 0.

[0058] The reflective surface **102d** is provided on the -Y side and the -Z side of the reflective surface **102b**, and the corner of the reflective surface **102d** on the +Y side and the +Z side is in contact with the corner of the reflective surface **102b**. The reflective surface **102d** inclines with respect to the reflective surface **102b** such that the corner of the reflective surface **102d** on the -Y side and the -Z side is closer to the virtual plane VP than the corner thereof on the +Y side and the +Z side. An angle between the reflective surface **102d** and the virtual plane VP corresponds to the incident angle of light to the incident surface **101**. An angle between the reflective surface **102d** and the reflective surface **102b** corresponds to the incident angle of light to the incident surface **101**. An inclination angle θ_{dh} of the reflective surface **102d** in the horizontal direction with respect to the reflective surface **102b** corresponds to an incident angle of light in the horizontal direction to be covered by the reflective surface **102d**. An inclination angle θ_{dv} of the reflective surface **102d** in the vertical direction with respect to the reflective surface **102b** corresponds to an incident angle of light in the vertical direction to be covered by the reflective surface **102d**. An absolute value of the inclination angle θ_{dv} in the vertical direction may be smaller than an absolute value of the inclination angle θ_{dh} in the horizontal direction.

[0059] The reflective surface **102d** is allotted incident light having a relatively large incident angle in the vertical direction on the positive side and a relatively large incident angle in the horizontal direction on the positive side. The angle between the reflective surface **102d** and the reflective surface **102b** may be about $\frac{1}{4}$ of the incident angle of the incident light to be covered by the reflective surface **102d**. If the optical member **100** is required to cover incident light having an incident angle of -10° or more and $+10^\circ$ or less in the vertical direction and an incident angle of -35° or more and $+35^\circ$ or less in the horizontal direction, the inclination angle θ_{dv} of the reflective surface **102d** in the vertical direction with respect to the reflective surface **102b** may be approximately 2.5° . The inclination angle θ_{dh} of the reflective surface **102d** in the horizontal direction with respect to the reflective surface **102b** may be approximately 10° .

[0060] The reflective surface **102e** is provided on the $-Y$ side of the reflective surface **102b** and is connected to the reflective surface **102b** at the side on the $+Y$ side. The reflective surface **102e** inclines with respect to the reflective surface **102b** such that the side on the $-Y$ side is closer to the virtual plane VP than the side on the $+Y$ side. An angle between the reflective surface **102e** and the virtual plane VP corresponds to the incident angle of light to the incident surface **101**. An angle between the reflective surface **102e** and the reflective surface **102b** corresponds to the incident angle of light to the incident surface **101**. An inclination angle θ_{eh} of the reflective surface **102e** in the horizontal direction with respect to the reflective surface **102b** corresponds to an incident angle of light in the horizontal direction to be covered by the reflective surface **102e**. An inclination angle θ_{ev} of the reflective surface **102e** in the vertical direction with respect to the reflective surface **102b** corresponds to an incident angle of light in the vertical direction to be covered by the reflective surface **102e**. An absolute value of the inclination angle θ_{ev} in the vertical direction may be smaller than an absolute value of the inclination angle θ_{eh} in the horizontal direction.

[0061] The reflective surface **102e** is allotted incident light having a relatively large incident angle in the horizontal direction on the positive side. The angle between the reflective surface **102e** and the reflective surface **102b** may be about $\frac{1}{4}$ of the incident angle of the incident light to be covered by the reflective surface **102e**. If the optical member **100** is required to cover incident light having an incident angle of -35° or more and $+35^\circ$ or less in the horizontal direction, the inclination angle θ_{ch} (refer to FIG. 7) of the reflective surface **102e** in the horizontal direction with respect to the reflective surface **102b** may be approximately 10° . The inclination angle θ_{ev} of the reflective surface **102e** in the vertical direction with respect to the reflective surface **102b** may be substantially 0.

[0062] The reflective surface **102f** is provided on the $-Y$ side and the $+Z$ side of the reflective surface **102b**, and the corner of the reflective surface **102f** on the $+Y$ side and the $-Z$ side is in contact with the corner of the reflective surface **102b**. The reflective surface **102f** inclines with respect to the reflective surface **102b** such that the corner of the reflective surface **102f** on the $-Y$ side and the $+Z$ side is closer to the virtual plane VP than the corner thereof on the $+Y$ side and the $-Z$ side. An angle between the reflective surface **102f** and the virtual plane VP corresponds to the incident angle of light to the incident surface **101**. An angle between the reflective surface **102f** and the reflective surface **102b** corresponds to the incident angle of light to the incident surface **101**. An inclination angle θ_{fh} of the reflective surface **102f** in the horizontal direction with respect to the reflective surface **102b** corresponds to an incident angle of light in the horizontal direction to be covered by the reflective surface **102f**. An inclination angle θ_{fv} of the reflective surface **102f** in the vertical direction with respect to the reflective surface **102b** corresponds to an incident angle of light in the vertical direction to be covered by the reflective surface **102f**. An absolute value of the inclination angle θ_{fv} in the vertical direction may be smaller than an absolute value of the inclination angle θ_{fh} in the horizontal direction.

[0063] The reflective surface **102f** is allotted incident light having a relatively large incident angle in the vertical direction on the negative side and a relatively large incident angle in the horizontal direction on the negative side. The angle between the reflective surface **102f** and the reflective surface **102b** may be about $\frac{1}{4}$ of the incident angle of the incident light to be covered by the reflective surface **102f**. If the optical member **100** is required to cover incident light having an incident angle of -10° or more and $+10^\circ$ or less in the vertical direction and an incident angle of -35° or more and $+35^\circ$ or less in the horizontal direction, the inclination angle θ_{fv} of the reflective surface **102f** in the vertical direction with respect to the reflective surface **102b** may be approximately -2.5° . The inclination angle θ_{fh} of the reflective surface **102f** in the horizontal direction with respect to the reflective surface **102b** may be approximately 10° .

[0064] The reflective surface **102g** is provided on the $+Y$ side and the $-Z$ side of the reflective surface **102b**, and the corner of the reflective surface **102g** on the $-Y$ side and the $+Z$ side is in contact with the corner of the reflective surface **102b**. The reflective surface **102g** inclines with

respect to the reflective surface **102b** such that the corner of the reflective surface **102g** on the +Y side and the -Z side is closer to the virtual plane VP than the corner thereof on the -Y side and the +Z side. An angle between the reflective surface **102g** and the virtual plane VP corresponds to the incident angle of light to the incident surface **101**. An angle between the reflective surface **102g** and the reflective surface **102b** corresponds to the incident angle of light to the incident surface **101**. An inclination angle θ_{gh} of the reflective surface **102g** in the horizontal direction with respect to the reflective surface **102b** corresponds to the incident angle of light in the horizontal direction to be covered by the reflective surface **102g**. An inclination angle θ_{gv} of the reflective surface **102g** in the vertical direction with respect to the reflective surface **102b** corresponds to an incident angle of light in the vertical direction to be covered by the reflective surface **102g**. An absolute value of the inclination angle θ_{gv} in the vertical direction may be smaller than an absolute value of the inclination angle θ_{gh} in the horizontal direction.

[0065] Each of the reflective surfaces **102g** is allotted incident light having a relatively large incident angle in the vertical direction on the positive side and a relatively large incident angle in the horizontal direction on the negative side. The angle between the reflective surface **102a** and the reflective surface **102b** may be about $\frac{1}{4}$ of the incident angle of the incident light to be covered by the reflective surface **102a**. If the optical member **100** is required to cover incident light having an incident angle of -10° or more and $+10^\circ$ or less in the vertical direction, the inclination angle θ_{gv} of the reflective surface **102g** in the vertical direction with respect to the reflective surface **102b** may be approximately 2.5° . The inclination angle θ_{gh} of the reflective surface **102g** in the horizontal direction with respect to the reflective surface **102b** may be approximately -10° .

[0066] The reflective surface **102h** is provided on the +Y side of the reflective surface **102b** and is connected to the reflective surface **102b** at the side on the -Y side. The reflective surface **102h** inclines with respect to the reflective surface **102b** such that the side of the reflective surface **102h** on the +Y side is closer to the virtual plane VP than the side thereof on the -Y side. An angle between the reflective surface **102h** and the virtual plane VP corresponds to the incident angle of light to the incident surface **101**. An angle between the reflective surface **102h** and the reflective surface **102b** corresponds to the incident angle of light to the incident surface **101**. An inclination angle θ_{hh} of the reflective surface **102h** in the horizontal direction with respect to the reflective surface **102b** corresponds to an incident angle of light in the horizontal direction to be covered by the reflective surface **102h**. An inclination angle θ_{hv} of the reflective surface **102h** in the vertical direction with respect to the reflective surface **102b** corresponds to an incident angle of light in the vertical direction to be covered by the reflective surface **102h**. An absolute value of the inclination angle θ_{hv} in the vertical direction may be smaller than an absolute value of the inclination angle θ_{hh} in the horizontal direction.

[0067] The reflective surface **102h** is allotted incident light having a relatively large incident angle in the horizontal direction on the negative side. The angle between the reflective surface **102h** and the reflective surface **102b** may be about $\frac{1}{4}$ of the incident angle of the incident light to be covered by the reflective surface **102h**. If the optical member **100** is required to cover incident light having an incident angle of -35° or more and $+35^\circ$ or less in the horizontal direction, the inclination angle θ_{hh} (refer to FIG. 7) of the reflective surface **102h** in the horizontal direction with respect to the reflective surface **102b** may be approximately -10° . The inclination angle θ_{hv} of the reflective surface **102h** in the vertical direction with respect to the reflective surface **102b** may be substantially 0.

[0068] The reflective surface **102i** is provided on the +Y side and the +Z side of the reflective surface **102b**, and the corner of the reflective surface **102i** on the -Y side and the -Z side is in contact with the corner of the reflective surface **102b**. The reflective surface **102i** inclines with respect to the reflective surface **102b** such that the corner of the reflective surface **102i** on the +Y side and the +Z side is closer to the virtual plane VP than the corner thereof on the -Y side and the -Z side. An angle between the reflective surface **102i** and the virtual plane VP corresponds to the

incident angle of light to the incident surface **101**. An angle between the reflective surface **102i** and the reflective surface **102b** corresponds to the incident angle of light to the incident surface **101**. An inclination angle θ_{ih} of the reflective surface **102i** in the horizontal direction with respect to the reflective surface **102b** corresponds to an incident angle of light in the horizontal direction to be covered by the reflective surface **102i**. An inclination angle θ_{iv} of the reflective surface **102i** in the vertical direction with respect to the reflective surface **102b** corresponds to an incident angle of light in the vertical direction to be covered by the reflective surface **102i**. An absolute value of the inclination angle θ_{iv} in the vertical direction may be smaller than an absolute value of the inclination angle θ_{ih} in the horizontal direction.

[0069] The reflective surface **102i** is allotted incident light having a relatively large incident angle in the vertical direction on the negative side and a relatively large incident angle in the horizontal direction on the positive side. The angle between the reflective surface **102i** and the reflective surface **102b** may be about $\frac{1}{4}$ of the incident angle of the incident light to be covered by the reflective surface **102i**. If the optical member **100** is required to cover incident light having an incident angle of -10° or more and $+10^\circ$ or less in the vertical direction and an incident angle of -35° or more and $+35^\circ$ or less in the horizontal direction, the inclination angle θ_{iv} of the reflective surface **102i** in the vertical direction with respect to the reflective surface **102b** may be approximately -2.5° . The inclination angle θ_{ih} of the reflective surface **102i** in the horizontal direction with respect to the reflective surface **102b** may be approximately -10° .

[0070] The reflective surface **102b** is substantially parallel to the virtual plane VP. Therefore, as indicated by a dotted arrow in FIG. 8, light LBb incident onto the incident surface **101** in the X direction is reflected by the reflective surface **102b** and is easily guided to the sensor **16**. However, assuming that light LBa incident onto the incident surface **101** from the direction inclining to the $-X$ side and the $+Z$ side with respect to the X direction is reflected by the virtual plane VP, the light LBa may be emitted in a direction deviating from the sensor **16**. Assuming that light Lbc incident onto the incident surface **101** from the direction inclining to the $-X$ side and the $-Z$ side with respect to the X direction is reflected by the virtual plane VP, the light Lbc may be emitted in a direction deviating from the sensor **16**.

[0071] On the other hand, the reflective surface **102a** inclines at the inclination angle θ_{av} (>0) in the vertical direction with respect to the virtual plane VP. Therefore, the light LBa incident onto the incident surface **101** from the direction inclining to the $-X$ side and the $+Z$ side in the X direction is reflected by the reflective surface **102a** and is easily guided to the sensor **16**, as indicated by an arrow of a one-dot chain line in FIG. 8.

[0072] The reflective surface **102c** inclines at the inclination angle θ_{cv} (<0) in the vertical direction with respect to the virtual plane VP. Therefore, the light Lbc incident onto the incident surface **101** from the direction inclining to the $-X$ side and the $-Z$ side in the X direction is reflected by the reflective surface **102c** and is easily guided to the sensor **16**, as indicated by a two-dot chain line arrow in FIG. 8.

[0073] As illustrated in FIG. 8, in addition to the reflective surface **102b** substantially parallel to the virtual plane VP, the reflective surface **102a** inclining to the positive side with respect to the reflective surface **102b** and the reflective surface **102c** inclining to the negative side with respect to the reflective surface **102b** are included in the reflective surface group **102**, thereby making it possible to readily enlarge an angle range in which light can be guided to the sensor **16** at the incident angle in the vertical direction.

[0074] When this is graphed, it is indicated by a solid line in FIG. 9. FIG. 9 is a diagram illustrating vertical angle dependency of sensor sensitivity. In FIG. 9, it can be seen that a fluctuation range ΔS_v of sensor sensitivity is relatively small and is suppressed within an allowable range in the angle range of -10° or more and $+10^\circ$ or less with respect to the incident angle in the vertical direction. It can be seen that -10° or more and $+10^\circ$ or less can be secured as the incident angle range in the vertical direction by a structure of the reflective surface group **102** illustrated in FIGS.

6 and 8.

[0075] The reflective surface **102b** is substantially parallel to the virtual plane VP. Therefore, as indicated by a dotted arrow in FIG. 10, the light LBb incident onto the incident surface **101** in the X direction is reflected by the reflective surface **102b** and is easily guided to the sensor **16**.

However, assuming that light LBe incident onto the incident surface **101** from a direction inclining to the -X side and the -Y side with respect to the X direction is reflected by the virtual plane VP, the light LBe may be emitted in a direction deviating from the sensor **16**. Assuming that light LBh incident onto the incident surface **101** from the direction inclining to the -X side and the +Y side with respect to the X direction is reflected by the virtual plane VP, the light LBh may be emitted in a direction deviating from the sensor **16**.

[0076] On the other hand, the reflective surface **102e** inclines at the inclination angle θ_{eh} (>0) in the vertical direction with respect to the virtual plane VP. Therefore, the light LBe incident onto the incident surface **101** from the direction inclining to the -X side and the -Y side with respect to the X direction is reflected by the reflective surface **102e** and is easily guided to the sensor **16**, as indicated by an arrow of a one-dot chain line in FIG. 10.

[0077] The reflective surface **102h** inclines at the inclination angle θ_{hh} (<0) in the horizontal direction with respect to the virtual plane VP. Therefore, the light LBh incident onto the incident surface **101** from the direction inclining to the -X side and the +Y side with respect to the X direction is reflected by the reflective surface **102h** and is easily guided to the sensor **16**, as indicated by a two-dot chain line arrow in FIG. 10.

[0078] As illustrated in FIG. 10, in addition to the reflective surface **102b** substantially parallel to the virtual plane VP, the reflective surface **102e** inclining to the positive side with respect to the reflective surface **102b** and the reflective surface **102h** inclining to the negative side with respect to the reflective surface **102b** are included in the reflective surface group **102**, thereby making it possible to readily enlarge an angle range in which light can be guided to the sensor **16** at the incident angle in the horizontal direction.

[0079] When this is graphed, it is indicated by a solid line in FIG. 11. FIG. 11 is a diagram illustrating horizontal angle dependency of sensor sensitivity. In FIG. 11, it can be seen that a fluctuation range ΔSh of the sensor sensitivity is relatively small and is suppressed within an allowable range in the angle range of -35° or more and $+35^\circ$ or less with respect to the incident angle in the horizontal direction. It can be seen that -35° or more and $+35^\circ$ or less can be secured as the incident angle range in the horizontal direction by a structure of the reflective surface group **102** illustrated in FIGS. 7 and 10.

[0080] As described above, in the present embodiment, the optical member **100** of the display device **1** includes the reflective surfaces **102a** to **102i** continuing to each other. The reflective surfaces **102a** to **102i** constitute a polygonal surface bulging out toward the opposite side of the incident side with respect to the virtual plane VP defined by the outer contours of the reflective surfaces **102a** to **102i**. With this structure, the optical member **100** can guide the incident light from the -X side to the sensor **16** in a wide angle range, and can expand the angle range in which the sensor **16** can detect the amount of light. Thus, it is possible to improve light guiding performance to the sensor **16** with respect to oblique incident light without using an expensive sensor.

[0081] It is noted that, in the reflective surface group **102** of the optical member **100**, some of the reflective surfaces **102a** to **102i** may have different areas. In one example, the area of the reflective surface **102b** provided at the center of the reflective surfaces **102a** to **102i** may be smaller than the areas of the other reflective surfaces **102a** and **102c** to **102i**. The intensity of the incident light having a relatively small incident angle is stronger than the intensity of the incident light having a relatively large incident angle. Therefore, by making the area of the reflective surface **102b** smaller than the areas of the other reflective surfaces **102a** and **102c** to **102i**, the intensities of light received by the reflective surfaces **102a** to **102i** can be made close to each other.

[0082] Alternatively, the reflective surface group of the optical member may be obtained by

dividing the virtual plane VP by $N \times M$ and deforming the virtual plane VP so as to bulge out toward the opposite side of the incident side while keeping a state where the $N \times M$ reflective surfaces are connected to each other. N is any integer of 2 or more. M is any integer of 2 or more. N and M may be the same as or different from each other. The values of N and M can be determined such that the areas of the respective divided reflective surface are equal to or larger than the area A1 of the light reception region 161 of the sensor 16. The values of N and M may be determined such that the areas of the respective divided reflective surfaces are equal to or larger than an area A2 (for example, 2.25 mm.^{sup.2}) obtained by adding a margin relating to an optical error to the area A1. In the embodiment, a configuration of N=3 and M=3 is exemplified.

[0083] In the case of, for example, N=2 and M=2, an optical member 200 may be configured as illustrated in FIGS. 12 to 14. FIG. 12 is a perspective view illustrating a configuration of an optical member 200 according to a modification of the embodiment. FIG. 13 is an XZ cross-sectional view illustrating the configuration of the optical member 200 according to the modification of the embodiment, and corresponds to an XZ enlarged cross-sectional view of the section B in FIG. 4 as viewed from the -Y direction. FIG. 14 is an XY cross-sectional view illustrating the configuration of the optical member 200 according to the modification of the embodiment, and illustrates a cross section of FIG. 13 taken along line D-D.

[0084] The optical member 200 includes a reflective surface group 202 instead of the reflective surface group 102 (FIGS. 5 to 7). The reflective surface group 202 includes reflective surfaces 202a to 202d continuing to each other. The reflective surfaces 202a to 202d constitute a polygonal surface bulging out toward the opposite side of the incident side with respect to the virtual plane VP. The virtual plane VP is defined by the outer contours of the reflective surfaces 202a to 202d (refer to FIG. 12). With this structure, the optical member 200 can guide the incident light from the -X side to the sensor 16 in a wide angle range, and can expand the angle range in which the sensor 16 can detect the amount of light. Each of the reflective surfaces 202a to 202d has a substantially triangular shape in plan view.

[0085] The reflective surfaces 202a to 202d are obtained by dividing the virtual plane VP by 2×2 and deforming the virtual plane VP so as to bulge out toward the opposite side of the incident side while keeping a state where 2×2 reflective surfaces are connected to each other. The reflective surfaces 202a to 202d are connected to each other via straight sides.

[0086] At this time, the reflective surfaces 202a to 202d incline at an angle to each other. Each of the reflective surfaces 202a to 202d extends obliquely to the virtual plane VP.

[0087] The reflective surfaces 202a to 202d may have different incident angles of incident light shared with each other. The reflective surface 202a is allotted incident light having a relatively small incident angle and incident light having a relatively large incident angle in the vertical direction on the positive side. The reflective surface 202b is allotted incident light having a relatively small incident angle and incident light having a relatively large incident angle in the vertical direction on the negative side. The reflective surface 202c is allotted incident light having a relatively small incident angle and incident light having a relatively large incident angle in the horizontal direction on the positive side. The reflective surface 202d is allotted incident light having a relatively small incident angle and incident light having a relatively large incident angle in the horizontal direction on the negative side.

[0088] The reflective surface 202a is provided on the -Z side of the reflective surfaces 202b to 202d. The reflective surface 202a is connected to the reflective surface 202c at the sides on the -Y side and the +Z side. The reflective surface 202a is connected to the reflective surface 202d at the sides on the +Y side and the +Z side. The reflective surface 202a has a corner on the +Z side in contact with a corner of the reflective surface 202b. The reflective surface 202a inclines with respect to the virtual plane VP such that the side on the -Z side is closer to the virtual plane VP than the angle of the +Z side. An angle between the reflective surface 202a and the virtual plane VP corresponds to the incident angle of light to the incident surface 101. An inclination angle θ_{ah200}

of the reflective surface **202a** in the horizontal direction with respect to the virtual plane VP corresponds to an incident angle of light in the horizontal direction to be covered by the reflective surface **202a**. An inclination angle θ_{av200} of the reflective surface **202a** in the vertical direction with respect to the virtual plane VP corresponds to an incident angle of light in the vertical direction to be covered by the reflective surface **202a**.

[0089] The reflective surface **202a** is allotted incident light having a relatively small incident angle and incident light having a relatively large incident angle in the vertical direction on the positive side. The angle between the reflective surface **202a** and the virtual plane VP may be about $\frac{1}{4}$ of the incident angle of the incident light to be covered by the reflective surface **202a**. If the optical member **200** is required to cover incident light having an incident angle of -10° or more and $+10^\circ$ or less in the vertical direction, the inclination angle θ_{av200} (refer to FIG. 13) of the reflective surface **202a** in the vertical direction with respect to the virtual plane VP may be approximately 2.5° . The inclination angle θ_{ah200} of the reflective surface **202a** in the horizontal direction with respect to the virtual plane VP may be substantially 0.

[0090] The reflective surface **202b** is provided on the +Z side of the reflective surfaces **202a**, **202c**, and **202d**. The reflective surface **202b** is connected to the reflective surface **202c** at the sides on the -Y side and the -Z side. The reflective surface **202b** is connected to the reflective surface **202d** at the sides on the +Y side and the -Z side. The reflective surface **202b** has a corner on the -Z side in contact with a corner of the reflective surface **202a**. The reflective surface **202b** inclines with respect to the virtual plane VP such that the side of the reflective surface **202b** on the +Z side is closer to the virtual plane VP than the corner thereof on the -Z side. An angle between the reflective surface **202b** and the virtual plane VP corresponds to the incident angle of light to the incident surface **101**. An inclination angle θ_{bh200} of the reflective surface **202b** in the horizontal direction with respect to the virtual plane VP corresponds to an incident angle of light in the horizontal direction to be covered by the reflective surface **202b**. An inclination angle θ_{bv200} of the reflective surface **202b** in the vertical direction with respect to the virtual plane VP corresponds to an incident angle of light in the vertical direction to be covered by the reflective surface **202b**.

[0091] The reflective surface **202b** is allotted incident light having a relatively small incident angle and incident light having a relatively large incident angle in the vertical direction on the negative side. The angle between the reflective surface **202b** and the virtual plane VP may be about $\frac{1}{4}$ of the incident angle of the incident light to be covered by the reflective surface **202b**. If the optical member **200** is required to cover incident light having an incident angle of -10° or more and $+10^\circ$ or less in the vertical direction, the inclination angle θ_{bv200} (refer to FIG. 13) of the reflective surface **202b** in the vertical direction with respect to the virtual plane VP may be approximately -2.5° . The inclination angle θ_{bh200} of the reflective surface **202b** in the horizontal direction with respect to the virtual plane VP may be substantially 0.

[0092] The reflective surface **202c** is provided on the -Y side of the reflective surfaces **202a**, **202b**, and **202d**. The reflective surface **202c** is connected to the reflective surface **202a** on the +Y side and the -Z side. The reflective surface **202c** is connected to the reflective surface **202b** on the +Y side and the +Z side. The reflective surface **202c** has a corner on the +Y side in contact with a corner of the reflective surface **202d**. The reflective surface **202c** inclines with respect to the virtual plane VP such that the side of the reflective surface **202c** on the -Y side is closer to the virtual plane VP than the corner thereof on the +Y side. An angle between the reflective surface **202c** and the virtual plane VP corresponds to the incident angle of light to the incident surface **101**. An inclination angle θ_{ch200} of the reflective surface **202c** in the horizontal direction with respect to the virtual plane VP corresponds to an incident angle of light in the horizontal direction to be covered by the reflective surface **202c**. An inclination angle θ_{cv200} of the reflective surface **202c** in the vertical direction with respect to the virtual plane VP corresponds to an incident angle of light in the vertical direction to be covered by the reflective surface **202c**.

[0093] The reflective surface **202c** is allotted incident light having a relatively small incident angle

and incident light having a relatively large incident angle in the horizontal direction on the positive side. The angle between the reflective surface **202c** and the virtual plane VP may be about $\frac{1}{4}$ of the incident angle of the incident light to be covered by the reflective surface **202c**. If the optical member **200** is required to cover incident light having an incident angle of -35° or more and $+35^\circ$ or less in the horizontal direction, the inclination angle θ_{ch200} (refer to FIG. 14) of the reflective surface **202c** in the horizontal direction with respect to the virtual plane VP may be approximately 10° . The inclination angle θ_{cv200} of the reflective surface **202c** in the vertical direction with respect to the virtual plane VP may be substantially 0.

[0094] The reflective surface **202d** is provided on the +Y side of the reflective surfaces **202a** to **202c**. The reflective surface **202d** is connected to the reflective surface **202a** at the sides on the -Y side and the -Z side. The reflective surface **202d** is connected to the reflective surface **202b** at the sides on the -Y side and the +Z side. The reflective surface **202d** has a corner on the -Y side in contact with a corner of the reflective surface **202c**. The reflective surface **202d** inclines with respect to the virtual plane VP such that the side of the reflective surface **202d** on the +Y side is closer to the virtual plane VP than the corner thereof on the -Y side. An angle between the reflective surface **202d** and the virtual plane VP corresponds to the incident angle of light to the incident surface **101**. An inclination angle θ_{dh200} of the reflective surface **202d** in the horizontal direction with respect to the virtual plane VP corresponds to an incident angle of light in the horizontal direction to be covered by the reflective surface **202d**. An inclination angle θ_{dv200} of the reflective surface **202d** in the vertical direction with respect to the virtual plane VP corresponds to the incident angle of light in the vertical direction to be covered by the reflective surface **202d**.

[0095] The reflective surface **202d** is allotted incident light having a relatively small incident angle and incident light having a relatively large incident angle in the horizontal direction on the negative side. The angle between the reflective surface **202d** and the virtual plane VP may be about $\frac{1}{4}$ of the incident angle of the incident light to be covered by the reflective surface **202d**. If the optical member **200** is required to cover incident light having an incident angle of -35° or more and $+35^\circ$ or less in the horizontal direction, the inclination angle θ_{dh200} (refer to FIG. 14) of the reflective surface **202d** in the horizontal direction with respect to the virtual plane VP may be approximately -10° . The inclination angle θ_{dv200} of the reflective surface **202d** in the vertical direction with respect to the virtual plane VP may be substantially 0.

[0096] In the above-described optical member **200** as well, the reflective surfaces **202a** to **202d** continuing to each other constitute a polygonal surface bulging out toward the opposite side of the incident side with respect to the virtual plane VP defined by the outer contours of the reflective surfaces **202a** to **202d**. With this structure, the optical member **200** can guide the incident light from the -X side to the sensor **16** in a wide angle range, and can expand the angle range in which the sensor **16** can detect the amount of light. Thus, it is possible to improve light guiding performance to the sensor **16** with respect to oblique incident light without using an expensive sensor.

[0097] While certain embodiments have been described, these embodiments have been presented by way of example only, and are not intended to limit the scope of the inventions. Indeed, the novel methods and systems described herein may be embodied in a variety of other forms; furthermore, various omissions, substitutions and changes in the form of the methods and systems described herein may be made without departing from the spirit of the inventions. The accompanying claims and their equivalents are intended to cover such forms or modifications as would fall within the scope and spirit of the inventions.

Claims

1. A display device comprising: a display panel; a transmission reflection member provided in front of the display panel; a sensor configured to detect an amount of light corresponding to incident

light incident onto the transmission reflection member; and an optical member provided near the transmission reflection member, the optical member including reflective surfaces continuing to each other, the optical member being configured to receive light, reflect the light by the reflective surfaces, and guide the reflected light to the sensor, the reflective surfaces constituting a polygonal surface bulging out toward an opposite side of an incident side with respect to a virtual plane defined by outer contours of the reflective surfaces.

2. The display device according to claim 1, wherein an area of each of the reflective surfaces is equal to or larger than an area of a light reception region of the sensor.

3. The display device according to claim 1, wherein each of the reflective surfaces extends along the virtual plane.

4. The display device according to claim 1, wherein the reflective surfaces include a first reflective surface extending along the virtual plane, and a second reflective surface extending obliquely to the virtual plane.

5. The display device according to claim 4, wherein the optical member further includes an incident surface on which light is incident, and an angle between the second reflective surface and the virtual plane corresponds to an incident angle of the light to the incident surface.

6. The display device according to claim 4, wherein the optical member further includes an incident surface on which light is incident, and an angle between the first reflective surface and the second reflective surface corresponds to an incident angle of the light to the incident surface.

7. The display device according to claim 4, wherein an area of the first reflective surface is smaller than an area of the second reflective surface.

8. An optical member comprising: an incident surface on which light is incident; an emitting surface extending from an end part of the incident surface in a direction intersecting the incident surface, the emitting surface facing a sensor serving to detect an amount of light; and reflective surfaces provided between the incident surface and the emitting surface in an optical path, the reflective surfaces continuing to each other, the reflective surfaces constituting a polygonal surface bulging out toward an opposite side of an incident side with respect to a virtual plane defined by outer contours of the reflective surfaces.

9. The optical member according to claim 8, wherein an area of each of the reflective surfaces is equal to or larger than an area of a light reception region of the sensor.

10. The optical member according to claim 8, wherein each of the reflective surfaces extends along the virtual plane.

11. The optical member according to claim 8, wherein the reflective surfaces include a first reflective surface extending along the virtual plane, and a second reflective surface extending obliquely to the virtual plane.

12. The optical member according to claim 11, wherein an angle between the second reflective surface and the virtual plane corresponds to an incident angle of the light to the incident surface.

13. The optical member according to claim 11, wherein an angle between the first reflective surface and the second reflective surface corresponds to an incident angle of the light to the incident surface.

14. The optical member according to claim 11, wherein an area of the first reflective surface is smaller than an area of the second reflective surface.
